

Harsh Bordekar

Simulation Engineer

Physics-Based Simulation, Model Verification & Validation, Uncertainty Quantification
PhD Candidate, DLR & TU Braunschweig — Aerospace Structural, Thermal & Materials Simulation

✉ harshbordekar@gmail.com

☎ +49 (0) 17673888310

🌐 [Harsh's LinkedIn](#)

📍 Braunschweig, Germany

🐙 github.com/harshbordekar-stack

🏠 Permanent Resident - Germany

PROFILE

Simulation and structural analysis engineer with 7+ years building, verifying, and maintaining physics-based simulation environments for real aerospace structures and vehicle hardware, spanning high-fidelity linear and non-linear FEM (ABAQUS, Nastran/Patran), coupled thermal-structural analysis, and multiscale, physics-informed damage modelling. Builds Monte Carlo and sensitivity-analysis capability into that work as standard practice, characterizing performance and probability of failure across the real range of operating conditions rather than a single nominal case. Validates models against real test and measurement data, component-level defect measurements, CT ground truth, externally and student-produced analysis, and treats model credibility, knowing where a model can be trusted and where it can't, as a first-class deliverable rather than an afterthought. Currently a PhD Candidate at DLR & TU Braunschweig, running large-scale simulation campaigns on HPC/SLURM infrastructure and building automation tooling that makes that simulation work faster and more accessible across analysis and student teams. Comfortable in Python, C++, and MATLAB, with experience supporting design trade studies and explaining simulation results and their limitations to engineers and non-specialist audiences alike.

Core Competencies: High-Fidelity Simulation Environment Development, Linear & Non-linear FEM (ABAQUS, Nastran/Patran), Thermal & Structural Simulation, Multiscale / Physics-Informed Modelling, Model Verification & Validation, Uncertainty Quantification & Sensitivity Analysis (Monte Carlo), Model Credibility & Sign-off, Simulation Tooling & Automation, Python, C++, MATLAB, HPC (SLURM), Cross-Disciplinary Collaboration

PROFESSIONAL EXPERIENCE

Structural Analysis / Simulation Engineer

DLR & TU Braunschweig

Physics-based simulation, model V&V & uncertainty quantification (PhD)

Feb 2021 - Present

- Built and maintained a probabilistic, physics-based multiscale simulation environment for leakage-onset prediction in CFRP hydrogen storage tanks, coupling microstructure-informed mesoscale damage models with stochastic spatial defect distributions to propagate failure across ply, laminate, and structural scales and estimate probability of failure.

Tools: Git, Docker, Python, ABAQUS, FEM, HPC/SLURM

- Developed Monte Carlo uncertainty quantification and sensitivity-analysis capability over the multiscale simulation outputs, generating leakage-probability-vs-strain relationships to characterize prediction confidence across the real range of strain/load conditions rather than a single nominal case, enabling risk-informed design decisions for next-generation CFRP pressure vessels.

Tools: Python (SciPy, NumPy), Monte Carlo methods, ABAQUS

- Built and ran large-scale thermal and structural simulation models for the Callisto reusable-rocket programme's metal upper housing bay, evaluating thermal protection effectiveness on HPC infrastructure to support design decisions for a real flight vehicle.

Tools: ABAQUS, SLURM (HPC), Python, Nastran, Patran

- Derived a certification-aligned leakage failure criterion (leakage envelope) mapping combined strain/load states to leakage onset, translating multiscale simulation outputs into structural sizing inputs and supporting design trade studies for pressure-vessel wall configuration under regulatory boundary conditions.

Tools: ABAQUS, Python, FEM post-processing using C++

- Verified and technically reviewed externally and student-produced simulation models, meshes, and results against defined methods and acceptance criteria, diagnosing failure modes traced to numerical, meshing, or data-quality issues and issuing documented sign-off-style findings on where a model was, and wasn't, trustworthy.

Tools: ABAQUS, Nastran, Python

- Built a simulation-tooling application that makes computational homogenization accessible without full-scale FEM runs: a physics-based CNN surrogate served through an agentic, tool-calling interface backed by a hierarchical RAG pipeline over 50 materials documents, with retrieval/prediction faithfulness validated at 93% against ground-truth homogenization results on a held-out set of more than 100 cases.

Tools: Python, Physics-Based CNN, Hierarchical RAG, LLM Agent Framework, Image Processing

- Instructed 60+ Master's students per year in FEM theory and ABAQUS-based CFRP modelling, covering Continuum Damage Mechanics (CDM), Progressive Damage Analysis (PDA), and Cohesive Zone Modelling (CZM), communicating simulation methods and their limitations to a technical audience.

Tools: ABAQUS, Automation using Python

Research Intern, Metallic Fatigue & Damage Tolerance

Fatigue-life / defect-tolerance assessment, validated against measured data

DLR-Braunschweig

Nov 2019 - Jan 2021

- Assessed fatigue and defect tolerance of aerospace-grade titanium (metallic) components using small-defect fatigue models (Murakami, El-Haddad), a physics-based simplification chosen deliberately over full-field simulation, to derive allowable defect sizes for a target fatigue life and link as-built defect populations to structural fatigue substantiation.

Tools: Small-defect fatigue models (Murakami, El-Haddad), Python (NumPy/SciPy)

- Coupled physical defect measurements with numerical simulation to establish a defect-tolerance qualification methodology, validating fracture-mechanics and fatigue-based predictions against measured defect populations and translating the result into allowable-defect acceptance limits for additively manufactured metallic parts.

Tools: FEM (ABAQUS), fracture mechanics, Python

- Developed a CT-based, explainable-AI defect-detection framework, validating classification output to **91%** accuracy (**89%** precision / **91%** recall) against ground-truth labels on **1200** CT scans of additively manufactured titanium parts, cutting manual review time to <0.5 sec per image while providing interpretable confidence outputs (SHAP, Grad-CAM, LIME) for engineering sign-off.

Tools: Python (scikit-learn / TensorFlow, SHAP / LIME / Grad-CAM), CT image processing

Student Research Assistant

Institute of Static and Dynamic

Leibniz University Hannover

Sep 2018 - Oct 2019

- Developed an ML framework for automated detection and quantification of fiber undulations in unidirectional CFRP laminates, generating spatially resolved defect metrics as direct inputs for probabilistic structural simulation models to assess manufacturing-induced variability on mechanical performance.

Tools: MATLAB, Python (scikit-learn / TensorFlow), image processing, NumPy/SciPy

Design Engineer

AK. Alloys

Gujarat Industrial Development Corporation

Mar 2017 - Mar 2018

- Designed sand-cast hardware components against manufacturing tolerances and process constraints, translating functional requirements into production-ready geometries compliant with draft angles, parting-line constraints, and dimensional quality standards.

Tools: CREO, SolidWorks, AutoCAD

- Managed witness casting and end-to-end quality assurance, verifying dimensional conformance and material integrity of cast hardware against engineering specifications and customer acceptance criteria.

Tools: CREO, SolidWorks, AutoCAD

SKILLS SUMMARY

Simulation & Modelling	FE Solvers (ABAQUS, Nastran/Patran) Linear & Non-linear FEM Thermal & Structural Simulation Multiscale / Physics-Informed Modelling Fracture Mechanics
Model Verification, Validation & Credibility	V&V Against Test / Measurement Data Sign-off Review of Externally Produced Models Ground-Truth Correlation Root-Cause Diagnosis (Numerical / Meshing Issues)
Uncertainty Quantification & Sensitivity Analysis	Monte Carlo Methods Probabilistic / Risk-Informed Assessment Small-Defect Fatigue Models (Murakami, El-Haddad) SALib
Programming & Scientific Computing	Python C++ MATLAB FORTRAN NumPy / SciPy / Pandas
HPC & Simulation Infrastructure	HPC / SLURM (Large-Scale Job Scheduling) Large-Scale Simulation Campaigns Simulation Automation & Tooling
Data Analysis & Visualization	Matplotlib Seaborn Plotly
ML & AI	Scikit-learn TensorFlow PyTorch Explainable AI (SHAP, Grad-CAM, LIME)
LLMs & Agentic / Tooling Systems	Retrieval-Augmented Generation (RAG) LangChain LangGraph Model Context Protocol (MCP) Tool-Calling
Version Control & Reproducibility	Git Docker
Application Development	VS Code JupyterLab Google Colab Streamlit Gradio
Languages	English (C1) German (B1)

AWARDS AND EDUCATION

 2nd Prize: Best Paper Award	DLR (German Aerospace Center) 2026
 3rd Prize: Best Paper Award	DLR (German Aerospace Center) 2024
PhD Candidate, Computational Methods in Engineering	DLR & TU Braunschweig, Germany Feb 2021 – 2026 (expected)
Dissertation: Physics-Informed, Multiscale Machine Learning for CFRP Composite Structures	
M.Sc. Computational Methods in Engineering	Leibniz University Hannover, Germany April 2018 – Dec 2020
Focus: Numerical Methods, Scientific Machine Learning, Deep Learning	

SELECTED PUBLICATIONS

1. **Bordekar, H.**, et al. “Microstructure-informed Probabilistic Multiscale Modelling of CFRP via Machine-Learning-Derived Stochastic Volume Elements.” *Manuscript in preparation*, 2026.
2. **Bordekar, H.**, et al. “MicroStructAgent: A Physics-Informed Agentic AI Framework for Computational Homogenization and Quality Assurance of Fiber Composite Laminates.” *Manuscript in preparation*, 2026.
3. **Bordekar, H.**, et al. “Microstructure Characterization of Fiber Composite Laminate Using eXplainable Artificial Intelligence.” *Journal of Composite Materials*, vol. 60, no. 1, 2026, pp. 3–25. [DOI](#)
4. **Bordekar, H.**, et al. “eXplainable Artificial Intelligence for Automatic Defect Detection in Additively Manufactured Parts Using CT Scan Analysis.” *Journal of Intelligent Manufacturing*, vol. 36, no. 2, 2025, pp. 957–974. [DOI](#)

REFERENCES

Prof. Dr. Christian Huehne	Abteilungsleiter Funktionsleichtbau. Deutsches Zentrum für Luft- und Raumfahrt. Institut für Systemleichtbau
Prof. Dr. Oliver Voelkerink	Head of Institute., IMA, TU-Braunschweig
Dr. Nicola Cersullo	R&T Project Manager @ Airbus Hamburg